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(1) Research Program/Project/Study Title: Production Performance, Heavy Metal Dynamics in Soil and Uptake of Selected Crops Applied with Different Forms of Compost-based Soil Amendment	
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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
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(7) Sector/Commodity Agriculture	(8) Discipline/Field of Specialization Crop, Soil and Environmental Sciences
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) Ensuring food security and safety through guaranteeing the quality of inputs that enter the crop production continuum necessitates the conduct of this investigation. This study aimed to determine the effect of compost-based soil amendment applied in various forms on the production performance, lead and cadmium dynamics in soil and uptake of selected crops. Field experiments were established on four selected food crops: radish, cucumber, sweet corn and kangkong applied with different forms of compost-based soil amendment namely: vermicast, vermi tea and ordinary compost following a Randomized Complete Block Design with three replications. Standard cultural management practices for the culture of each crop were followed. Growth and yield data were gathered and analyzed statistically. Soil and plant tissue samples were collected and analyzed to ascertain the levels of lead (Pb) and cadmium (Cd). Results revealed that better growth and yield of radish, cucumber, sweet corn and kangkong were attained with the application of vermicast. Application of solid form of compost-based soil amendment namely vermicast and ordinary compost favored net accumulation of Pb in soil grown with radish and cucumber, while vermi tea application favored net uptake. There was net accumulation of Pb in soil grown to sweet corn regardless of compost form applied. Kangkong proved to be a hyperaccumulator plant in terms of Pb uptake regardless of the compost form used.	
(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) cadmium, compost, crop yield, lead, vermi tea, vermicast	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Composting is the practice of creating humus-like organic materials outside of the soil by mixing, piling, or otherwise storing organic materials under conditions conducive to aerobic decomposition and nutrient conservation. The decomposition processes and organisms involved are very similar to those already described for the formation of humus in soils. The main difference is that with composting, decay occurs outside of the soil, and in such a concentrated fashion as to generate considerable heat. The finished product, compost, is popular as mulch, as an ingredient for potting mixes, and as an organic soil conditioner and slow-release fertilizer (US-EPA, 2023). Vermicomposting or earthworm composting is a non-thermophilic process (Edwards and Burrows, 1988 as cited by Joshi et al., 2014), which involves a joint action of earthworms and mesophilic microorganisms (Aira et al., 2002) for the conversion of organic wastes into a valuable end product called vermicompost. It has been reported as a viable, cost effective technique for the efficient management of organic solid waste (Longsdon, 1994; Hands et al., 1998 as cited by Shobit et al., 2015). As its two useful products, earthworm biomass can further be processed into proteins as a source of animal feeds (Hartstein and Harstein, 1981), while vermicompost contains many plant growth hormones, soil enzymes and essential plant nutrients (Ndegwa and Thompson, 2001). However, several studies as reported by Morgan and Morgan (1999) and (Wani, et al., 2013) have found that earthworms effectively bioaccumulate or biodegrade several organic and inorganic chemicals including 'heavy metals', 'organochlorine pesticide' and 'polycyclic aromatic hydrocarbons' (PAHs) residues, and in the remediation of heavy metals (Sahu and Sharma, 2016; Pattnaik and Reddy, 2010). Vermitea is a lateral product of vermicomposting process which contains nitrogen, phosphorous, many micronutrients, and hormones among others. It also contains earthworm enzyme	

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and promote plant growth and yield and increase resistance of plant to disease and pests. It has also dissolved nutrient, some organic acids and earthworm mucus (Singh et al., 2011). Vermicompost tea has influential effect on improving plant vigor, nutritive quality and yield of crop (Tara, 2003; Chen, 2006).

Concerning food security and safety, guaranteeing the quality and safety of compost-based soil amendments which enters the soil-plant continuum are important areas for research and development, ensuring the freedom from heavy metal hazard while sustaining the production of organic crops. It is in this premise that Capiz State University proposed the approval of this research and development project which cut across the foregoing areas of concern.

This study specifically aimed to (1) determine the effect of compost-based soil amendment applied in various forms on the production performance of selected crops; and (2) evaluate the effect of compost-based soil amendment applied in various forms on the lead and cadmium dynamics in soil and uptake of selected crops.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental Method

This study was composed of four set-up corresponding to the different selected crops: C1 – Kangkong (leafy crop), C2 – Radish (tuberous vegetable), C3 – Cucumber (fruit vegetable), and C4 - Sweet Corn (green crop). Each set-up was tested for the different forms of compost-based soil amendment, namely: F1 – Vermicast, F2- Vermi tea, and F3 –Ordinary Compost. The study was laid out using the Randomized Complete Block Design with three replications.

Cultural Practices

Land Preparation. Thirty six plots measuring 2.5 m x 3 m were prepared to accommodate the four crops and three treatments, each replicated three times. The plots were enclosed with cemented hollow blocks. Each replication or blocking was spaced 1m while distance between plot was 0.5 m. Border plots with a width of 1 m were provided. The area was prepared by thorough cultivation with an interval of one week per operation. Plastic cover was later installed in all blocks to protect crops from damage brought about by unfavourable weather conditions

Securing the seeds. The seeds of open-pollinated variety of the different crops were bought from reputable agricultural supplier in the region.

Sowing of seeds. The seeds of the different crops were sown in seed boxes containing garden soil, compost, and sand at equal proportion. Seeds were sown in prepared seed box by evenly spreading them in a finger made furrow.

Pricking of seedlings. As soon as the seeds germinated or when the radicle protruded, the over crowded germinated seeds were transferred to the seedling trays to provide them more space to grow and avoid root intertwining. The seedlings were kept in the nursery until they produced three true leaves, and with vigorous appearance before they were transplanted in the field.

Hardening of seedlings. Seedlings were hardened as soon as they produced true leaves, by gradually exposing the seedlings to open sunlight to avoid transplanting shock during transplanting. This was done by exposing the seedlings to open sunlight starting early in the morning until 9:00 a.m. on the first week. The number of exposures to sunlight was increased in the following weeks until they produced three or four true leaves and ready for transplanting.

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Transplanting the seedlings. The seedlings with three to four true leaves, vigorous, and disease-free were transplanted in the fields following the recommended planting distance. Transplanting was done late in the afternoon in order to minimize transplanting shock.

Care of seedlings. The transplanted seedlings were initially covered with banana bracts for four days during extreme sunlight to prevent them from wilting. Watering of plant using sprinkler was done every morning and late in the afternoon.

Cultivation and weeding. The cultivation of space between rows and hills was done two weeks after transplanting. Shallow cultivation was practiced using hand trowel to avoid mechanical injury of the roots. Hand weeding was done by pulling the weeds with the use of a trowel as soon as the weeds appeared in the plots.

Watering. The plants were watered whenever necessary especially when there is no rain.

Application of different forms of compost-based soil amendment. Vermicast and ordinary compost were applied to the designated experimental plots on the final stage of land preparation. The rate of application was determined based on the recommended rate. For the vermi tea, application is done two weeks before transplanting and every week thereafter except during the flowering stage. The concentration used is 20 ml/liter of water. The rate of application per plot is four liters and the amount is increased by 1 liter every two weeks.

Pest and disease management, and control. Cleanliness and sanitation were strictly implemented in the project. Pest and disease monitoring was strictly done. In case of high pest and disease incidence which threatened the performance of plants, botanical, OHN and organically formulated pesticides were used. Insect repellent plants were also planted around the experimental area. The population and kinds of pest that occurred were observed.

Harvesting/Priming. For the leafy and tuberous vegetables, the crops were harvested as soon as they manifested the maturity indices. For the fruit vegetables, they were harvested when they reached the marketable stage. This stage was gauged by the manifestation of the maturity indices. For the sweet corn, it was harvested 65 to 70 days after transplanting.

Data Gathering. Data gathered varied depending on the crop. For the leafy vegetables, the following data were gathered: plant height, stem girth, number, length, and width of leaves, biomass, weight of marketable parts, and plot yield. For the fruit vegetables, the following data were gathered: plant height, stem girth, number of marketable and non-marketable fruits, weight of marketable and non-marketable fruits, biomass, length and diameter of fruit, and plot yield. For tuberous vegetables, the following data were gathered: plant height, length of tuber, diameter of tuber, biomass, weight of marketable and non-marketable parts, and plot yield.

The initial concentration of Pb and Cd of soil in the experimental area was taken. The soil was also sampled at the end of every cropping season to determine the changes in Pb and Cd level. The different economically important part of the plant were subjected to plant tissue analysis to determine the heavy metal (Pb and Cd) uptake. Collection of plant samples for analysis was done on the third or fourth priming which was the peak of harvest for the fruit vegetables, while for the crops harvested at one time, plant samples were taken at harvest.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

RADISH

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Growth parameters of radish such as plant height, number of leaves and biomass were comparable with respect to the forms of compost-based soil amendment applied (Table 1.0). Plant height of radish ranged from 28.51 cm to 31.01 cm, number of leaves ranged from around 15 to 17, while biomass ranged from 226.00 g to 363.67 g. On the other hand, the width of radish leaves was significantly affected by the form of compost-based soil amendment applied, wherein application of vermicast resulted to wider leaves of radish as compared to the application of vermi tea and ordinary compost.

Table 1.0. Means of different growth parameters of radish applied with different forms of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Plant Height (^{ns})	Number of Leaves (^{ns})	Width of Leaves (*)	Biomass (^{ns})
Vermicast	31.01	16.80	10.82a	363.67
Vermi tea	28.51	16.23	9.95b	282.67
Ordinary Compost	29.30	14.70	9.38b	226.00

In terms of the yield parameters of radish, the application of different form of compost-based soil amendment resulted to comparable length and diameter of tuber, weight of marketable and non-marketable tubers and plot yield (Table 2.0). The length of tuber ranged from 20.06 cm to 23.73 cm, diameter of tuber ranged from 3.70 cm to 4.31 cm, weight of marketable tubers ranged from 157.00 g to 257.67 g, weight of non-marketable tubers ranged from 68.33 g to 102.50 g, while plot yield ranged from 13.77 kg to 19.26 kg.

Table 2.0. Means of different yield parameters of radish applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Length of Tuber (^{ns})	Diameter of Tuber (^{ns})	Weight of Marketable Tubers (^{ns})	Weight of Non- marketable Tubers (^{ns})	Plot Yield (^{ns})
Vermicast	23.73	4.31	257.67	102.50	19.26
Vermi tea	22.84	3.94	200.83	75.50	15.28
Ordinary Compost	20.06	3.70	157.00	68.33	13.77

When it comes to the lead content of soil grown to radish (Table 3.0), the application of different form of compost-based soil amendment such as vermicast and ordinary compost resulted to an increasing trend from the start of planting up to the harvest of the second cropping. This denotes that the area was contaminated with lead and there was a net accumulation of lead in the soil when solid form of compost-based soil amendment was applied. On the other hand, the application of vermi tea to radish resulted to a decline in total lead content after the first cropping but an increase in the lead content of the soil after harvest of the second cropping. This could be attributed to the liquid nature of the applied form of compost-based soil amendment where lead becomes more soluble thus prone to uptake than retention.

Table 3.0. Lead content (ppm) of soil grown to radish applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicast	32.60	55.07	82.03
Vermi tea	50.23	31.33	68.30
Ordinary compost	25.53	43.67	78.13

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In the case of cadmium content of soil grown to radish (Table 4.0), there was no cadmium detected in all soil samples with respect to the form of compost-based soil amendment applied. This implies that the area is not contaminated with cadmium. Also, this result is probably due to the trace amount of cadmium in the vermicast, vermi tea and ordinary compost applied which was not also detected after each cropping season.

Table 4.0. Cadmium content (ppm) of soil grown to radish applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicast	Not detected in all samples		
Vermi tea			
Ordinary compost			

CUCUMBER

All growth parameters of cucumber namely plant height, stem girth and biomass were comparable with respect to the form of compost-based soil amendment applied (Table 5.0). Plant height of cucumber ranged from 240.72 cm to 320.72 cm, stem girth ranged from 0.67 cm to 0.84 cm, while biomass ranged from 205.07 g to 265.63 g.

Table 5.0. Means of different growth parameters of cucumber applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Plant Height (^{ns})	Stem Girth (^{ns})	Biomass (^{ns})
Vermicast	320.72	0.84	265.63
Vermi tea	240.72	0.67	214.97
Ordinary Compost	285.34	0.75	205.07

In terms of the yield parameters of cucumber (Table 6.0), the application of different form of compost-based soil amendment resulted to comparable length of fruits, weight of marketable and non-marketable fruits and plot yield. The length of fruits ranged from 18.58 cm to 20.16 cm, weight of marketable fruits ranged from 152.77 g to 232.88 g, weight of non-marketable fruits ranged from 13.62 g to 16.20 g, while plot yield ranged from 15.56 kg to 21.81 kg. On the other hand, the diameter of fruits was significantly affected by the form of compost-based soil amendment applied, wherein application of vermicast resulted to bigger fruits as compared to the application of vermi tea and ordinary compost.

Table 6.0. Means of different yield parameters of cucumber applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Length of Fruits (^{ns})	Diameter of Fruits (**)	Weight of Marketable Fruits (^{ns})	Weight of Non-marketable Fruits (^{ns})	Plot Yield (^{ns})
Vermicast	20.16	4.98a	232.88	16.20	21.81
Vermi tea	18.58	4.29b	163.16	13.62	15.56
Ordinary Compost	18.58	4.41b	152.77	14.57	16.61

When it comes to the lead content of soil grown to cucumber (Table 7.0), data revealed that the area was contaminated with lead. Likewise, the application of different form of compost-based soil amendment such as vermicast, vermi tea and ordinary compost resulted to an increasing trend from

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the start of planting up to the harvest of the first cropping. This was followed by a declining trend in all lead content values from the first cropping to the second cropping.

Table 7.0. Lead content (ppm) of soil grown to cucumber applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicast	35.47	106.80	78.73
Vermi tea	37.23	100.53	53.40
Ordinary compost	33.80	111.47	49.73

In the case of cadmium content of soil grown to cucumber (Table 8.0), there was no cadmium detected in all soil samples with respect to the form of compost-based soil amendment applied. This implies that the area is not contaminated with cadmium. Also, this result is probably due to the trace amount of cadmium in the vermicast, vermi tea and ordinary compost applied.

Table 8.0. Cadmium content (ppm) of soil grown to cucumber applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicast	Not detected in all samples		
Vermi tea			
Ordinary compost			

SWEET CORN

Growth parameters of sweet corn such as plant height, width of ear, number of leaves and stem girth were comparable with respect to the form of compost-based soil amendment applied (Table 9.0). Plant height of sweet corn ranged from 169.25 cm to 191.75 cm, width of ear ranged from 4.61 cm to 5.19 cm, number of leaves ranged from 9.70 to 10.77, while stem girth ranged from 1.39 cm to 1.63 cm. On the other hand, the length of ear and biomass of sweet corn were significantly affected by the form of compost-based soil amendment applied, wherein the application of vermicast resulted to longer corn ears and heavier sweet corn plants as compared to the application of vermi tea and ordinary compost. Vermicast or vermicompost is better than ordinary compost due to its higher nitrogen, phosphorus, and potassium content and ability to improve the soil structure and increase water-holding capacity (Singh, 2020). Vermicast or vermicompost is better and highly recommended because it contains most macronutrients, and its nutrient composition is often more significant than the regular garden soil. Vermicomposting increases soil fertility and water resistance and helps in germination, plant growth, and crop yield. They nurture the soil with plant growth hormones such as auxins, gibberellic acid, etc. (Iderawumi, 2022). In the study of Allardice (2014), he found that vermicompost contained higher mineral nutrients, increased microbial populations and activity in the soil, and increased ryegrass biomass compared to thermophilic compost.

Table 9.0. Means of different growth parameters of sweet corn applied with different compost forms.

Form of Compost-Based Soil Amendment	Plant Height (ns)	Length of Ear (*)	Width of Ear (ns)	No. of Leaves (ns)	Stem Girth (ns)	Biomass (**)
Vermicast	191.75	14.67 ^a	5.19	10.77	1.63	554.67 ^a
Vermi tea	169.25	12.43 ^b	4.61	9.90	1.46	393.43 ^b

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Ordinary Compost	176.08	12.17 ^b	4.84	9.70	1.39	397.17 ^b
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In terms of the yield parameters of sweet corn (Table 10.0), the application of different form of compost-based soil amendment resulted to significantly different weight of ear and weight of marketable parts, but comparable in terms of the weight on non-marketable parts and plot yield. Heavier weight of corn ears and marketable parts were obtained from the application of vermicast as compared to the application of vermi tea and ordinary compost. On the other hand, the weight of non-marketable corn parts ranged from 20.0 g to 29.0 g, while plot yield ranged from 3.40 kg to 4.62 kg.

Several researchers worldwide have found that the nutrient profile in vermicompost is generally higher than in traditional compost. Vermicompost can enhance soil fertility physically, chemically, and biologically. Physically, vermicompost-treated soil has better aeration, porosity, bulk density, and water retention. Chemical properties such as pH, electrical conductivity, and organic matter are also improved for better crop yield. Nevertheless, enhanced plant growth could not be satisfactorily explained by improvements in the nutrient content of the soil, which means that other plant growth-influencing materials are available in vermicompost (Lim et al., 2015). Vermicompost applications also increased total organic carbon, total N, P, K, Ca, Zn, and Mn contents, and decreased soil pH. Soil density and porosity also improved when vermicompost was added to the soil (Azarmi et al., 2008).

Table 10.0. Means of different yield parameters of sweet corn applied with different compost forms.

Form of Compost-Based Soil Amendment	Weight of Ear (**)	Weight of Marketable Parts (**)	Weight of Non-Marketable Parts (^{ns})	Plot Yield (^{ns})
Vermicast	264.83 ^a	234.00 ^a	29.00	4.62
Vermi tea	190.63 ^b	167.00 ^b	21.00	3.40
Ordinary Compost	196.83 ^b	176.00 ^b	20.00	3.88

When it comes to the lead content of soil grown to sweet corn (Table 11.0), data revealed that the area was contaminated with lead. Likewise, the application of different compost forms such as vermicast and vermi tea resulted to an increasing trend from the start of planting up to the harvest of the first cropping. This was followed by a declining trend in the lead content values from the first cropping to the second cropping in plots applied with vermicast and vermi tea. On the other hand, a continued increasing trend from planting up to the second cropping in the lead content of soil was detected in plots of sweet corn applied with ordinary compost. This may imply that under successive sweet corn production, application of ordinary compost may prove effective in the retention of lead in the soil.

Table 11.0. Lead content (ppm) of soil grown to sweetcorn applied with different forms of compost.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicompost	39.33	55.40	45.90
Vermi tea	32.37	48.93	48.70
Ordinary compost	38.97	46.57	51.40

In the case of cadmium content of soil grown to sweet corn (Table 12.0), there was no cadmium detected in all soil samples with respect to the form of compost applied. This implies that the area is not contaminated with cadmium. Also, this result is probably due to the trace amount of cadmium in the vermicast, vermi tea and ordinary compost applied.

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Table 12.0. Cadmium content (ppm) of soil grown to sweetcorn applied with different forms of compost.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicompost	Not detected in all samples		
Vermi tea			
Ordinary compost			

KANGKONG

Growth parameters of kangkong such as plant height, stem girth and biomass were comparable with respect to the form of compost-based soil amendment applied (Table 13.0). Plant height of kangkong ranged from 56.23 cm to 59.66 cm, stem girth ranged from 0.67 cm to 0.75 cm, while biomass ranged from 51.80 g to 67.80 g. On the other hand, the number of leaves of kangkong was significantly affected by the form of compost-based soil amendment applied, wherein the application of vermi tea resulted to more kangkong leaves as compared to the application of vermicast and ordinary compost. This could be due to the liquid nature of vermi tea wherein nutrients become more soluble which facilitated better uptake which benefited the growth of kangkong being a leafy vegetable.

Table 13.0. Means of different growth parameters of kangkong applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Plant Height ^(ns)	Stem Girth ^(ns)	Number of Leaves ^(*)	Biomass ^(ns)
Vermicast	59.66	0.75	10.30b	67.80
Vermi tea	56.23	0.67	11.53a	51.80
Ordinary Compost	58.83	0.69	9.77b	67.77

In terms of the yield parameters of kangkong (Table 14.0), the application of different form of compost-based soil amendment resulted to comparable length and width of leaves, and weight of marketable parts. The length of leaves ranged from 18.19 cm to 18.48 cm, width of leaves ranged from 3.28 cm to 4.42 cm, while weight of marketable parts ranged from 45.00 g to 62.00 g. However, plot yield of kangkong was significantly affected by the form of compost-based soil amendment applied, wherein the application of vermicast and ordinary compost comparably resulted to higher plot yield as compared to the application of vermi tea. This could be due to the nature of vermicast and ordinary compost which contained more nutrients and their slow-release nature may have sustained the nutrient demand of kangkong across different ratoons. On the other hand, vermi tea being liquid though provide more soluble form of nutrient but may not be sustainable in providing the nutrient requirement of kangkong for several ratoons.

Table 14.0. Means of different yield parameters of kangkong applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	Length of Leaves ^(ns)	Width of Leaves ^(ns)	Weight of Marketable Parts ^(ns)	Plot Yield ^(*)
Vermicast	18.48	4.42	62.00	16.66a
Vermi tea	18.19	3.28	45.00	10.15b
Ordinary Compost	18.29	3.91	61.17	14.25a

When it comes to the lead content of soil grown to kangkong (Table 15.0), data revealed that the area was contaminated with lead. Likewise, the application of different form of compost-based soil amendment such as vermicast and vermi tea resulted to a declining trend from the start of planting up

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to the harvest of the second cropping. In the case of ordinary compost application, it resulted to a decline in the lead content of soil from planting up to the harvest of the first cropping but increase during the second cropping. This may imply that under successive kangkong production, application of vermicast and vermi tea may not be ideal as less retention of lead took place in the soil, and more lead may have been taken up.

Table 15.0. Lead content (ppm) of soil grown to kangkong applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicompost	73.57	43.6	29.90
Vermi tea	69.00	44.0	40.30
Ordinary compost	77.73	35.1	38.30

In the case of cadmium content of soil grown to kangkong (Table 16.0), there was no cadmium detected in all soil samples with respect to the form of compost-based soil amendment applied. This implies that the area is not contaminated with cadmium. Also, this result is probably due to the trace amount of cadmium in the vermicast, vermi tea and ordinary compost applied.

Table 16.0. Cadmium content (ppm) of soil grown to kangkong applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	SAMPLING TIME		
	Initial	After First Cropping	After Second Cropping
Vermicompost	Not detected in all samples		
Vermi tea			
Ordinary compost			

Heavy Metal Uptake

Concerning the heavy metal detected in plant tissues (Table 17.0), uptake of lead was evident in kangkong regardless of the form of compost-based soil amendment applied. This may imply that indeed, kangkong is a potent lead hyperaccumulator plant. Likewise, it was worthy to observe that application of vermi tea resulted to the uptake of lead in radish and cucumber but not in sweet corn. This could be attributed to the liquid form of vermi tea that makes nutrients including heavy metal more soluble thus readily become available for uptake. The nature of sweet corn being tall-statured and robust plant may have allocated lead in most of its parts thus becomes diluted and was not traceable during tissue analysis.

Table 17.0. Lead content (ppm) in plant tissues applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	CROP			
	Kangkong	Radish	Cucumber	Sweetcorn
Vermicompost	36.13	ND	ND	ND
Vermi tea	179.42	62.00	26.20	ND
Ordinary compost	48.00	ND	ND	ND

*ND = Not detected

On the other hand, cadmium was not detected in all tissue samples of kangkong, radish, cucumber and sweet corn when applied with different form of compost-based soil amendment (Table 18.0). This is probably due to the trace amount of cadmium in the vermicast, vermi tea and ordinary compost applied which made them untraceable in plant parts.

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Table 18.0. Cadmium content (ppm) in plant tissues applied with different form of compost-based soil amendment.

Form of Compost-Based Soil Amendment	CROP			
	Kangkong	Radish	Cucumber	Sweetcorn
Vermicompost	Not detected in all samples			
Vermi tea				
Ordinary compost				

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Based from the findings of this study, the following conclusions are drawn:

- * Better growth and yield of radish, cucumber, sweet corn and kangkong were attained with the application of vermicast.
- * Application of solid form of compost-based soil amendment namely vermicast and ordinary compost favored net accumulation of lead in soil grown with radish and cucumber, while net uptake was favored with the application of vermi tea.
- * There was a net accumulation of lead in soil grown to sweet corn regardless of the form of compost-based soil amendment applied.
- * Kangkong proved to be a hyperaccumulator plant in terms of lead uptake regardless of the form of compost-based soil amendment applied.

The following recommendations are forwarded:

- * Application of vermicast is suggested for increased soil organic matter and better growth and yield of radish, cucumber, sweet corn and kangkong when grown in Pb-contaminated soil.
- * Application of solid form of compost-based soil amendment namely vermicast and ordinary compost is recommended over that of liquid form (vermi tea) when growing radish, cucumber and sweet corn in Pb-contaminated soil, as solid form favors retention while liquid form facilitates uptake.
- * In Pb-contaminated soil, kangkong should not be grown as the crop is proven as hyperaccumulator of lead.
- * Related studies be conducted to verify the findings of this present investigation.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Quality and safe organically-produced food crops

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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